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GUILDFORD HIGH SCHOOL

ENOKI CALVERT-ANSARI
ecalvert-ansari@guildfordhigh.co.uk

BIOMECHANICAL AND DESIGN IMPLICATIONS OF REPLACING INTEGRATED CARBON-FIBRE PLATES WITH REMOVABLE RUNNING SHOE INSERTS

Research Proposal submitted to the International Academic Research Competition (IARCO) 2026

Biomechanical Engineering:

Category: Junior · Year 11 ·

GUILDFORD, ENGLAND
2026

1 INTRODUCTION

Running has traditionally been a very accessible sport due to its simplicity and minimal equipment requirements [1]. However, high-performance “super shoes” combining responsive foam with embedded carbon fibre plates and aggressive forward-curving rocker shape, have dramatically changed the landscape of the sport [2]. Research shows that advanced footwear can improve running economy and performance, making these shoes appealing for athletes on all levels [3]. However, premium carbon-plated shoes typically cost between £160 - £300 or more, meaning many athletes cannot afford to buy the latest model each season, creating an uneven playing field [4].

Although carbon-plated shoes can improve running performance, their fixed design means they are not always adaptable to every runner’s biomechanics, foot shape or training needs [5]. For younger or recreational runners, the rigidity, rocker shape and highly responsive design may reduce the loading of calves and Achilles tendon, placing unfamiliar stress on the forefoot, leading to injuries like stress fractures and plantar fasciitis [6]. Therefore, a fully plated, aggressive racing shoe may not be suitable for every athlete if used too often without gradual adaptation. Runners need a more adaptable way to explore its benefits while still considering comfort, training purpose, biomechanics and injury risk [7].

In addition, running shoes have become consumable items as companies prioritise lightweight, maximum energy return materials which can reduce the usable lifespan of the shoe [8]. Also, the rigid carbon plate puts intense, localized pressure on the surrounding soft foam during every stride, speeding up this material fatigue [9]. Athletic footwear creates significant environmental waste, with around 33 million pairs of running shoes sent to landfill each year subsequently producing synthetic foams, plastics, rubbers and adhesives that take approximately 1,000 years to decompose [10]. Replacing an entire shoe when only its performance characteristics have deteriorated therefore creates unnecessary material waste as well as additional cost for athletes.

Carbon-plated running shoes, designed for elite running, have become desirable at an amateur level despite their high cost, limited durability and suitability issues, creating financial waste and problems regarding sustainability and fairness in amateur running. Therefore, this research will investigate whether a removable carbon fibre insert, if shaped ergonomically and combined with a cushioning layer, could maintain benefits associated with carbon-plated running shoes while improving affordability, adaptability and product longevity.

2 LITERATURE REVIEW

This research is important because research has demonstrated that advanced footwear can reduce metabolic demand by approximately 2–3%, creating a measurable competitive advantage [11]. A recent UK study found that 30% of parents surveyed were unable to afford the equipment and

clothing their children required for sport, while 37% reported that their children had missed sporting opportunities because of cost [12].

Another limitation of existing carbon-plated footwear is that its mechanical properties are largely fixed, despite there being no single optimal carbon-plated design for every runner. Individual improvements in running economy have ranged from negligible or even negative responses to improvements approaching 10%, demonstrating substantial variation in the interaction between runner and shoe [13]. Poorly matched designs may also have biomechanical consequences as carbon-plated footwear alters foot and ankle mechanics, including metatarsophalangeal movement and loading distribution [14].

Environmental research provides further justification for reducing whole-shoe replacement. Life-cycle analysis estimated that a pair of synthetic running shoes produces approximately 14 ± 2.7 kg CO_2-equivalent, with around 97% arising from material processing and manufacturing [15]. Performance can also deteriorate before a shoe is physically unusable. Furthermore, running footwear combines foams, rubbers, textiles and adhesives, making separation and recycling difficult [16]. Extending the useful life of an existing shoe or replacing only a performance component could therefore reduce material consumption and manufacturing demand.

Current approaches address affordability, adaptability and sustainability separately rather than as a single problem. Affordability is mainly addressed through different tiers of running footwear, including lower-cost alternatives to premium racing shoes. Adaptability is increasingly being addressed through footwear-selection systems and emerging customised shoe designs based on the individual runner's biomechanics [17]. Sustainability approaches currently focus on reducing manufacturing emissions and developing recycling or material-recovery systems [18, 19]. A recent sports-shoe prototype containing 62–68% bio-based material achieved an estimated carbon footprint of 7.8–8.6 kg CO_2-eq per pair, compared with approximately 14 ± 2.7 kg CO_2-eq previously calculated for a conventional synthetic running shoe [20, 21]. These approaches demonstrate progress, but there remains no established solution that addresses all problems whilst retaining the performance benefits of carbon-plated footwear.

3 METHODOLOGY

This study will use a mixed-method experimental research design combining laboratory testing, human performance testing and finite element analysis (FEA). I will design and manufacture a removable carbon-fibre plate. For the prototype, quantitative data will be collected; mass, thickness, longitudinal bending stiffness and durability. Human testing will collect quantitative data on speed and heart rate, alongside qualitative feedback on comfort, stability, pressure and perceived propulsion. A FEA will compare the mechanical behaviour of an integrated carbon-fibre plate with a removable plate under identical simulated loading conditions. Pressure distribution, energy return , plate bending and stress/strain distribution, maximum pressure concentration and long term fatigue resistance will be analysed. Figure 1 the project stages.

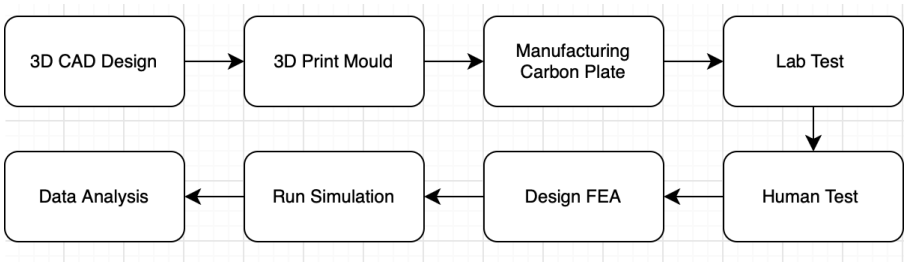


Figure 1: Flow diagram showing a synopsis of my plan

Three footwear conditions will be tested: (1) A conventional non-carbon-plated running shoe; (2) The same shoe with the removable carbon-fibre prototype inserted; and (3) A commercially available running shoe with an integrated carbon-fibre plate. The independent variable is therefore the type of footwear/plate system, while the dependent variables include mechanical stiffness, running time and speed, average heart rate, distance covered, comfort ratings and FEA outputs such as plate deformation, pressure and stress distribution.

3.1 Design & Material Selection

Initial designs will combine analysis of existing carbon-plated shoes (Figure 2) with anthropometric data from a specific participant. The material selection of carbon fibre offers the best balance of low weight, stiffness, strength and thinness, allowing it to support propulsion without adding significant mass or bulk to the shoe [22, 23]. The chosen manufacturing method is carbon-fibre cloth with epoxy resin, shaped using a 3D printed mould and wet lay-up process, to create a curved rocker design. Although carbon fibre is expensive and difficult to recycle, it is strong, durable and increasingly recyclable through improved recovery methods, so it remains the most appropriate material for this prototype [24, 25].

a) b) c)

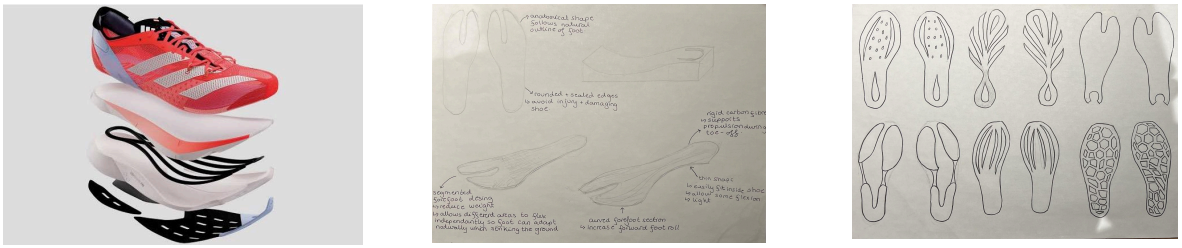


Figure 2: (a) Adidas shoe including carbon-fibre ENERGYRODS, (b) Annotated final design, (c) Initial design idea generation

The finite element analysis computer simulation will involve research on carbon-plated running shoes, finite element modelling of footwear, midsole foam behaviour and carbon-fibre plate stiffness will be used to guide the design of the simulation. Furthermore, this study will use a quantitative computational research design. I will create 3D geometries of the midsole foam layers, carbon plate, insole and upper material, as in Figure 2. I will use CAD software to create two main shoe models, one with an integrated carbon-plate embedded within the midsole foam and one with a removable carbon-plate insert, positioned above the midsole and closer to the foot, with a cushioning layer used to reduce direct pressure.

The CAD geometries will be imported into finite element software such as ANSYS or FEBio. The same mesh settings and assumptions will be applied consistently between models. The material models will be applied to the relevant structures and layers in the FEA software. Boundary conditions (e.g. velocity of the foot) will be defined to simulate the mechanical loading that occurs during running, particularly during forefoot loading and toe-off.

The same loading conditions will be applied to both models. This is important because the aim is not to recreate every detail of a full running stride, but to compare how the two plate positions behave. A further cyclic or repeated-loading simulation may be carried out to estimate fatigue-related behaviour. This would not provide a complete prediction of long-term durability, but it could help identify areas where repeated bending may cause high stress concentration or potential failure.

Lab Testing

First, the length, width and thickness of the carbon plate will be measured using digital callipers. Measurements will be taken at the heel, midfoot and forefoot to check whether the plate has a consistent thickness. The plate will be weighed using digital scales and recorded. Longitudinal bending stiffness will then be measured using a three-point bend test. The plate will be placed across two supports at a fixed distance apart, and a known load will be applied to the centre. The deflection will be measured and recorded. This test is important because it shows how much the plate resists bending during toe-off, which is one of the key performance-related functions of a carbon plate. The test will be repeated three times and an average result will be calculated to improve reliability. Fatigue resistance will also be tested by repeatedly bending the plate 100–500 times using a controlled manual method. It will then be inspected for cracking, splitting, delamination or permanent deformation.

Human Testing

This human testing study will use a mixed-methods approach to assess whether the carbon-fibre plate improves running economics whilst maintaining comfort and stability. The protocol is divided into two phases: (1) Performance test, the athlete will complete three 100 metre runs at maximum effort; and (2) endurance-efficiency test, the athlete will complete a 10-minute Zone 2 run in each footwear condition. After each trial, the shoe and plate will be inspected. The insole will be removed to check whether the plate has shifted position, caused pressure marks, damaged the shoe or shown signs of cracking, splitting or delamination. To ensure accuracy and participant safety, trials incorporate adequate recovery time to prevent fatigue, and footwear is rigorously inspected after each run for structural damage, pressure marks, or plate shifting. Human testing will only take place after the plate has passed the lab tests. The plate must have smooth, sealed edges and should be used with a cushioning layer above it. The tester will be instructed to stop immediately if they feel pain, instability, rubbing, sharp pressure or discomfort. All testers will sign a consent form and their PII will not be stored.

5 EXPECTED RESULTS

The final physical product will be a 2mm thick, removable carbon-fibre running shoe plate made using carbon-fibre cloth and epoxy resin. I predict during the FEA both carbon-plate models to show similar bending behaviour, but the removable plate may create higher local pressure because it sits closer to the foot, while the integrated plate should distribute force more evenly through the midsole foam. This will identify cushioning and plate geometry as important areas for further optimisation. For the laboratory testing, I expect the removable carbon-fibre plate to show high longitudinal bending stiffness and a Young's modulus of 50–70 GPa. After 100–500 repeated bending cycles, there should be no visible cracking, delamination or permanent deformation after repeated bending. In the 100 m sprint test, I expect the RCP to outperform the NPS but remain slightly below the ICP. In the 100 m test, the RCP should reach 25 km/h, with both plated shoes averaging under 15 seconds and the ICP potentially 1–3% faster. During the 10-minute Zone 2 run, the athlete should maintain 70–80% maximum heart rate (150–175 bpm) and cover approximately 2.5 km in both plated shoes. The ICP is expected to provide greater comfort, stability and efficiency, while the RCP should feel more propulsive than the NPS.

6 POTENTIAL LIMITATIONS

Potential limitations include manufacturing only one pair of removable carbon-fibre plates. If they fail during testing, the experimental programme may not be able to continue. To reduce this risk. A further limitation is that the prototype will be designed around the anthropometric measurements of one specific athlete, meaning that the initial human-testing results will not represent a wide range of runners. If the first stage of testing is successful, a second participant, ideally a female runner could be tested.

7 CONCLUSION

Carbon-plated running shoes can improve performance, but their high cost, fixed design and limited durability create problems of affordability, adaptability and sustainability, particularly when runners must replace an entire shoe to access or maintain the benefits of the carbon-fibre plate. This research will investigate whether a removable carbon-fibre insert could address these issues by providing a lower-cost performance component that can be adapted to an individual runner and transferred or replaced without discarding the whole shoe.

By comparing conventional, removable-plate and integrated carbon-plated footwear through laboratory testing, human testing and FEA, the study will assess whether a removable system can retain useful biomechanical benefits while identifying limitations in areas such as pressure distribution, comfort, stiffness and durability. If successful, the research could demonstrate the feasibility of separating the carbon-fibre performance element from the shoe itself and provide a basis for future development of personalised plate geometry, cushioning across a wider range of runners.

REFERENCES

[1] R. Hitchings and A. Latham, "How 'social' is recreational running? Findings from a qualitative study in London and implications for public health promotion," Health & Place, vol. 46, pp. 337–343, Dec. 2016, doi: 10.1016/j.healthplace.2016.10.003.

[2] K. Hébert-Losier and M. Pamment, "Advancements in running shoe technology and their effects on running economy and performance – a current concepts overview," Sports Biomechanics, vol. 22, no. 3, pp. 1–16, Aug. 2022, doi: 10.1080/14763141.2022.2110512.

[3] W. Hoogkamer, S. Kipp, J. H. Frank, E. M. Farina, G. Luo, and R. Kram, "A Comparison of the Energetic Cost of Running in Marathon Racing Shoes," Sports Medicine, vol. 48, no. 4, pp. 1009–1019, Nov. 2017, doi: 10.1007/s40279-017-0811-2.

[4] B. Dyer, "A Pragmatic Approach to Resolving Technological Unfairness: the Case of Nike's Vaporfly and Alphafly Running Footwear," Sports Medicine - Open, vol. 6, no. 1, May 2020, doi: 10.1186/s40798-020-00250-1.

[5] F. Fu et al., "Effect of the Construction of Carbon Fiber Plate Insert to Midsole on Running Performance," Materials, vol. 14, no. 18, p. 5156, Jan. 2021, doi: 10.3390/ma14185156.

[6] PGC1 Coaching, "The Drawbacks of Using Carbon-Plated Running Shoes in Training – A Running Coach's View," 13 Feb. 2025. [Online]. Available: https://www.pgc1coaching.com/post/the-drawbacks-of-using-carbon-plated-running-shoes-in-training-a-running-coach-s-view

[7] P. Mai et al., "Towards functionally individualised designed footwear recommendation for overuse injury prevention: a scoping review," BMC Sports Science, Medicine and Rehabilitation, vol. 15, no. 1, Nov. 2023, doi: 10.1186/s13102-023-00760-x.

[8] R. Verdejo and N. J. Mills, "Heel–shoe interactions and the durability of EVA foam running-shoe midsoles," Journal of Biomechanics, vol. 37, no. 9, pp. 1379–1386, Sept. 2004, doi: 10.1016/j.jbiomech.2003.12.022.

[9] C. Sánchez, "The Ultimate Guide to Running Shoe Foams," RunRepeat, updated 9 Oct. 2025. [Online]. Available: https://runrepeat.com/uk/guides/running-shoe-foams-guide

[10] Aspire PR, "JogOn: One Year On and 15,000 Running Shoes Removed from Landfill," Jul. 2023. [Online]. Available: https://www.aspirepr.co.uk/news/jogon-one-year-on-and-15000-running-shoes-removed-from-landfill

[11] E. N. Kobayashi et al., "Metabolic effects of carbon-plated running shoes: A systematic review and meta-analysis," Frontiers in Sports and Active Living, 2026.

[12] "Sport is NOT a luxury: The perceived impact of the cost-of-living crisis on sport and physical activity opportunities for children and adolescents in Northern Ireland," 2025.

[13] "Biomechanical factors associated with intraindividual differences in running economy across advanced footwear technology models in long-distance runners," 2026.

[14] A. J. Staikos and S. Rahimifard, "An air-based automated material recycling system for postconsumer footwear products," Resources, Conservation and Recycling, vol. 69, pp. 90–99, 2012.

[15] L. Cheah et al., "Manufacturing-focused emissions reductions in footwear production," Journal of Cleaner Production, vol. 44, pp. 18–29, 2013.

[16] M. J. Lee and S. Rahimifard, "An air-based automated material recycling system for postconsumer footwear products," Resources, Conservation and Recycling, vol. 69, pp. 90–99, 2012, doi: 10.1016/j.resconrec.2012.09.008

[17] S. Müller, P. Helwig, H.-H. Dickhuth, and F. O. Mayer, "Clinical effectiveness of customised sport shoe orthoses for overuse injuries in runners: A randomised controlled study," British Journal of Sports Medicine, vol. 45, no. 12, pp. 959–965

[18] S. A. D. Dissanayake et al., "Life cycle and end-of-life management options in the footwear industry: A review," Journal of Cleaner Production, 2020. The review identifies mixed-material construction and recovery difficulties as major barriers to footwear recycling.

[19] T. Loganathan et al., "Design and development of sustainable sports shoes using recycled materials in compliance with testing standards," Discover Applied Sciences, vol. 8, Art. no. 993, 2026.

[20] T. Loganathan et al., "Design and development of sustainable sports shoes using recycled materials in compliance with testing standards," Discover Applied Sciences, vol. 8, Art. no. 993, 2026.

[21] L. Cheah et al., "Manufacturing-focused emissions reductions in footwear production," Journal of Cleaner Production, vol.

[22] K. A. Alharbi *et al.*, "Comprehensive Review of the Properties and Modifications of Carbon Fiber-Reinforced Thermoplastic Composites," *Polymers*, vol. 13, no. 15, 2021.

[23] "Metabolic effects of carbon-plated running shoes: a systematic review and meta-analysis," *Sports Medicine / PMC*, 2026.

[24] P. Alam, D. Mamalis, C. Robert, C. Floreani, and C. M. Ó Brádaigh, "The fatigue of carbon fibre reinforced plastics — A review," *Composites Part B: Engineering*, vol. 166, pp. 555–579, 2019.

[25] M. Yuan, Z. Li, and Z. Teng, "Progress and prospects of recycling technology for carbon fiber reinforced polymer," *Frontiers in Materials*, vol. 11, 2024.